Tidal potential measured on a coastal–inland optical-clock baseline

Introduction

Optical clocks now reach fractional systematic uncertainties near 10^{-19}. Separate lattice-clock experiments have achieved instabilities of a few $10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is averaging time.[1,2] Clock networks extend these measurements beyond individual laboratories.[3,4,5] Optical fibres connect terrestrial sites, while long-distance free-space experiments are testing the transfer capability needed for future ground-to-space links.[6,7] These connections support optical timekeeping and a future redefinition of the second, and allow comparisons of gravitational potential between distant sites.[8,9]

The relative ticking rates of clocks depend on gravitational potential. Near Earth's surface, a fractional frequency change of 10^{-18} corresponds to about 1 cm of equivalent height.[9] Experiments have resolved gravitational redshifts on laboratory scales and measured static potential differences between distant sites.[10,11,12,13] Measuring how this potential changes, and separating the processes responsible, places further demands on a clock network. Periodic signals with similar frequencies require sufficient phase coverage to be distinguished, whereas irregular changes require the necessary precision during the event itself. Interruptions can also leave the frequency reference level uncertain between observing runs. The signals a network can detect therefore depend on its sensitivity, the duration of observations and how continuously it operates.

Tides provide a predictable signal against which to test this capability. The Moon and Sun change Earth's gravitational potential through their tidal fields, deformation of the solid Earth and redistribution of ocean water. At individual sites, solid-Earth tidal potentials can reach amplitudes of several $\mathrm{m}^2/\mathrm{s}^2$. Smaller effects from pole tides and non-tidal mass changes in the atmosphere, oceans and continental water can matter at the 10^{-18} fractional-frequency level.[14] Solar heating also drives atmospheric pressure tides that force the ocean, while polar motion generates a slower pole tide through changes in centrifugal forcing.[15,16] Beyond these periodic signals, satellite gravity has revealed mass redistribution associated with the changing water cycle and climate.[17,18] Repeated clock comparisons between fixed sites could help investigate these changes.[19] The measured signal is a difference between sites, so its amplitude and time dependence vary with the baseline and the physical process.

Here we detect tidal variations in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan. The clocks are separated by 692 km and connected by a 1,350-km fibre link.[20] The combined solid-Earth and ocean-loading prediction gives an M_2 fractional-frequency amplitude of 4.78×10^{-18}, equivalent to about 4.4 cm of height.[16,21] We use 545.9 hours of observations spread over 353 days. The astronomical phase is known at each observation time, even when the record is interrupted. We therefore allow each observing segment its own frequency offset and combine its measured variations with those from other dates. The changing relative phase of the lunar and solar tides permits joint estimation of the M_2+N_2 and neighbouring S_2+K_2 responses, with phases and relative amplitudes fixed within each group.

References

1. Arnold, K. J. et al. Lu^+ optical frequency references with accuracy verified at the 19th digit. *Nature* **658**, 60–64 (2026). https://doi.org/10.1038/s41586-026-11072-8
2. Oelker, E. et al. Demonstration of 4.8×10^{-17} stability at 1 s for two independent optical clocks. *Nat. Photon.* **13**, 714–719 (2019). https://doi.org/10.1038/s41566-019-0493-4
3. Boulder Atomic Clock Optical Network (BACON) Collaboration. Frequency ratio measurements at 18-digit accuracy using an optical clock network. *Nature* **591**, 564–569 (2021). https://doi.org/10.1038/s41586-021-03253-4
4. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nat. Commun.* **7**, 12443 (2016). https://doi.org/10.1038/ncomms12443
5. Lindvall, T. et al. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica* **12**, 843–852 (2025). https://doi.org/10.1364/OPTICA.561754
6. Shen, Q. et al. Free-space dissemination of time and frequency with 10^{-19} instability over 113 km. *Nature* **610**, 661–666 (2022). https://doi.org/10.1038/s41586-022-05228-5
7. Caldwell, E. D. et al. Quantum-limited optical time transfer for future geosynchronous links. *Nature* **618**, 721–726 (2023). https://doi.org/10.1038/s41586-023-06032-5
8. Dimarcq, N. et al. Roadmap towards the redefinition of the second. *Metrologia* **61**, 012001 (2024). https://doi.org/10.1088/1681-7575/ad17d2
9. Mehlstäubler, T. E., Grosche, G., Lisdat, C., Schmidt, P. O. & Denker, H. Atomic clocks for geodesy. *Rep. Prog. Phys.* **81**, 064401 (2018). https://doi.org/10.1088/1361-6633/aab409
10. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical clocks and relativity. *Science* **329**, 1630–1633 (2010). https://doi.org/10.1126/science.1192720
11. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). https://doi.org/10.1038/s41586-021-04349-7
12. Takano, T. et al. Geopotential measurements with synchronously linked optical lattice clocks. *Nat. Photon.* **10**, 662–666 (2016). https://doi.org/10.1038/nphoton.2016.159
13. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nat. Phys.* **14**, 437–441 (2018). https://doi.org/10.1038/s41567-017-0042-3
14. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). https://doi.org/10.1088/0026-1394/53/6/1365
15. Arbic, B. K. Atmospheric forcing of the oceanic semidiurnal tide. *Geophys. Res. Lett.* **32**, L02610 (2005). https://doi.org/10.1029/2004GL021668
16. Petit, G. & Luzum, B. (eds) *IERS Conventions (2010).* IERS Technical Note **36** (2010). https://iers-conventions.obspm.fr/content/tn36.pdf
17. Tapley, B. D., Bettadpur, S., Ries, J. C., Thompson, P. F. & Watkins, M. M. GRACE measurements of mass variability in the Earth system. *Science* **305**, 503–505 (2004). https://doi.org/10.1126/science.1099192
18. Rodell, M. et al. Emerging trends in global freshwater availability. *Nature* **557**, 651–659 (2018). https://doi.org/10.1038/s41586-018-0123-1
19. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of

optical clock networks for validating satellite-derived time-variable gravity data. *Geophys. J. Int.* **226**, 764–779 (2021). https://doi.org/10.1093/gji/ggab132
20. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. Preprint at https://arxiv.org/abs/2610.34250v1 (2026).
21. CNES. FES2022 (Finite Element Solution) Tidal model (Version 2024). https://doi.org/10.24400/527896/A01-2024.004 (2024).